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Meritocracy and the Networks It Forged: China's Imperial Examination
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ABSTRACT

Institutions that claim to reward merit often also serve as engines that drive the formation of important social networks. We examine this dual role in China's Imperial Examination, one of history's longest-lived meritocratic systems. We draw on exam performance data for 24,269 individuals across 112 national examinations in the Qing dynasty (1644–1905) and trace who among them rose to become top officials. Strong exam performance increased the likelihood of attaining high office, but advancement also depended on individuals' ties to powerful examiners. Because performance and ties to powerful examiners were strongly complementary, individuals' success appeared to rest on merit even as social networks played a role. Following the major political transformations of the 1840s, the importance of examiner ties persisted, although the influential examiners themselves changed. These findings help explain why such systems retain legitimacy even when networks matter greatly.

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1 Introduction

Across societies, institutions that claim to allocate opportunities on the basis of merit often become entangled with more informal social networks. Competitive entry exams, elite schools, and bureaucratic selection systems are designed to channel talent into positions of authority. Yet, alongside their meritocratic function, these institutions frequently cultivate relationships—between mentors and protégés or among peers—that exert a powerful influence on the trajectory of individual careers. From Oxford and Cambridge, where alumni networks have long shaped Britain's political class (Useem, 1986), to France's Grandes Écoles, which dominate state administration (Bourdieu, 1998), to Ivy League universities in the United States, where alumni ties structure professional advancement (e.g., Collins, 1979; Rivera, 2016; Markovits, 2019), meritocratic institutions routinely double as engines of social networks. The coexistence of merit and networks raises fundamental questions about how elites are produced and whether social networks undermine the very premise of meritocracy itself. Although this interplay has long been recognized across disciplines, systematic empirical evidence on its workings and evolution remains scarce.

China's imperial examination system is a useful case study for exploring this dynamic. Operating for more than a millennium—the earliest form dating to 605 CE—it was a highly competitive civil service system designed to select government officials based on individual merit rather than birth or connections. Revolutionary for its time, the examination tested candidates on a wide range of subjects, including classical Chinese literature and Confucian texts, and offered even those of humble origins the possibility of attaining high office. Beyond its role in bureaucratic recruitment, the examination system also fostered lasting social networks, particularly between examiners and examinees. The Qing dynasty (1644–1905) was renowned for its rich archival records, which provide researchers today an unusually well-defined setting to study both merit and networks within the same institution.

Beyond functioning as a mechanism to identify talent, the examination halls also served as fertile ground for social network formation. Examiners and examinees could forge enduring ties, and fellow candidates, who were screened and selected for political promise by the same examiners, often became lifelong allies. In particular, the examiner–examinee bond, often described as a master–disciple relationship, could prove decisive for the career advancement of successful candidates. In our analysis, we focus on the contrast and interaction between exam performance, the key metric of merit in this setting, and the examiner–examinee ties forged through the examination institution. In doing so, our work speaks to broader skepticism about meritocracy, generally.

Empirically, we use data on 24,269 candidates from 112 national-level exams (Zhu and Xie, 1980), known as *Jinshi*, in the Qing dynasty (1644–1905). Passing these exams made candidates eligible for top office, defined as rank three or above in the nine-rank system—positions with substantial authority over national policy. Yet only a small share ultimately reached this level, about 12.6 percent. Our central question is how merit and networks jointly shaped the probability of reaching this elite tier and how their interaction varied across candidates and political contexts.

We measure merit using exam performance, defined as a candidate's rank within an exam cohort of roughly 240 successful candidates. We also consider a salient institutional feature that divided candidates into a high and a low category at a cutoff, analogous to elite college thresholds in some contemporary education systems. Our main performance measure is percentile rank, defined relative to this cutoff between high and low categories, which allows us to test for any discontinuity generated by the institution. While exam performance may partly reflect family background, we treat it as the institution's own proxy for merit and ask whether it operated as claimed. We also use information on one's hometown to gauge whether omitted personal background is crucial for our findings.

To capture the role of exam networks, we study how the post-exam career trajectories of examiners shaped the advancement of their examinees. Each exam cohort was overseen by multiple examiners whose subsequent power varied in both degree and duration. We construct a continuous measure of ties to powerful examiners by calculating, for each candidate, the total number of years that their examiners held key central positions within 10 years after the exam (with alternative windows of 5 and 20 years). Intuitively, this approach compares candidates whose examiners later became central figures—thus able to provide an early career boost—with peers from other cohorts who had fewer such ties.

Our study comprises three sets of analyses. Our first shows that both merit and networks mattered for becoming a top official. On average, a 10 percentile improvement in exam performance is associated with a 3 percentage point (24 percent) increase in the probability of becoming a top official. The institutional categorization also mattered: There is a jump in the chance of becoming a top official for those who belonged to the better-performing group, and the performance–success slope is also higher for them. These patterns support the relevance of the exam institutions: Not only exam performance but also the institution that classified candidates affected career outcomes.

An important limitation of our data is the lack of direct measures of family background, which may be correlated with exam performance. Nevertheless, when we control for hometown

characteristics or include hometown fixed effects, our baseline results remain virtually unchanged. This finding suggests that broad differences in family background across places are unlikely to fully account for the observed relationship between exam performance and career outcomes, although we cannot rule out the role of within-hometown variation in family background.

Ties with powerful examiners (hereafter, "examiner power") were also consequential. After examiners themselves advanced to key central positions, their former examinees became more likely to reach top office. Quantitatively, a one-standard-deviation increase in ties to powerful examiners is associated with a 17 percent increase in the probability of reaching top office.

To shed light on whether examiner power reflects more than mentoring or human capital transmission, we exploit variation in the timing of examiner power. When we include both pre-exam and post-exam examiner power in the same specification, post-exam power shows the stronger association with candidates' later advancement. This finding suggests that an examiner's later position of influence may have mattered beyond any fixed qualities the examiner brought to the mentoring process, although mentoring and human capital transmission may still have operated alongside such influence.

Moreover, one potential concern is reverse causality: Successful examinees may have advanced in tandem with, or even contributed to, their examiners' rise, leading to a comovement in career trajectories. To assess this possibility, we divide examiners into two groups: those on a rising trajectory and those whose power remained stable or declined. If reverse causality were a major force, one might expect the estimated effect of examiner power to be stronger among the former group. In the data, however, the estimated association is similar across the two groups, suggesting that reverse causality is unlikely to fully explain our findings.

We also examine how the importance of merit and examiner power evolves over the course of an examinee's career. Analyzing the timing of appointments to top office (e.g., within 10 years, 11–15 years, 16–20 years of obtaining the degree), we find that compared with exam performance, examiner power is especially influential early in an examinee's career. In other words, when the overall likelihood of attaining top office was still low, examiner power provided a particularly valuable boost.

Our second set of analyses examines the interplay between exam performance and examiner power. Are these two complements, substitutes, or independent factors in shaping advancement? The answer sheds light on the degree of meritocracy within the system. Empirically, we find strong complementarity: Candidates with better performance benefited disproportionately from examiner

power, while ties were useful but less effective for weaker performers.

To assess the welfare implications of this complementarity, one would need to identify the underlying channels through which ties operated—whether primarily through patronage or information—and evaluate whether connected candidates performed better or worse. Such an analysis lies beyond the scope of this article.

Nevertheless, the observed complementarity pattern carries two important implications. On the one hand, it depicts a system more meritocratic than a counterfactual system in which networks crowd out merit; thus, the influence of connections does not necessarily negate the existence of meritocracy. On the other hand, because those who advanced were typically both better performing and well connected, their success could appear to rest purely on merit, even when networks played a decisive role. In this sense, the complementarity between performance and connections lent a veneer of legitimacy to the meritocratic ideal, masking the underlying power of social ties.

Our final analysis examines how the relative importance of merit and ties to powerful examiners evolved over time. A key advantage of our long historical data, spanning more than 260 years, is that it allows us to observe how these forces shifted or persisted across dramatically different political environments. In our context, a crucial turning point came in the late 1830s and early 1840s, when China faced a series of profound crises: wars with Western powers (1840s–1890s), the devastating Taiping Rebellion (1851–1864), and widespread political turmoil that ultimately culminated in the fall of the dynasty in 1911. These shocks fundamentally challenged the stability of the imperial system. We therefore ask: Did such disruptions alter the balance between merit and networks in shaping official advancement?

We find that the influence of both exam performance and examiner power persisted through these disruptions. What changed was not the importance of networks per se, but whose networks mattered: After 1840, ties to Han examiners became more valuable while ties to Manchu examiners declined in influence, consistent with the dynasty's growing reliance on Han elites in its later decades. These results highlight the resilience of network-based advancement, even in the face of profound political turbulence, and reveal how shifting patterns of elite reliance reshaped which networks conferred advantage.

Our article connects two strands of thinking on meritocracy. A classic sociological tradition emphasizes that mobility systems may combine open competition with elite sponsorship (Turner, 1960). More recent critiques argue that modern meritocratic institutions can reproduce inequality while preserving the appearance of fairness (e.g., Rivera, 2016; Markovits, 2019; Sandel, 2021;

Wooldridge, 2021). Our findings link these perspectives by showing how a celebrated meritocratic institution could simultaneously function as a competitive selection device and as a generator of career-enhancing ties.

Skepticism toward meritocracy generally falls into two broad categories. The first holds that "merit" itself reflects inherited advantages, most notably family background. While our data do not include detailed family information, we find that exam performance cannot be accounted for by candidates' hometown characteristics. This pattern may be particular to our historical setting, where family investment in education could be less decisive than in modern contexts.

The second critique views merit as a veneer for social networks. Our findings speak more directly to this concern. The imperial examination system represents a striking case of a celebrated meritocratic institution that simultaneously served as a key arena for elite network formation. Social ties clearly mattered for career advancement. Yet, because performance and connections were complements rather than substitutes, the system retained a degree of meritocratic character. In this sense, meritocracy is not a binary condition but a continuum, and its legitimacy often rests on the subtle coexistence of merit and networks.

Our study is also related to a broad literature on selection processes in which merit and networks interact, including in bureaucratic promotion (Jia, Kudamatsu and Seim, 2015; Xu, 2018), research funding (Li, 2017), and academic publishing (Brogaard, Engelberg and Parsons, 2014). Some studies document substitution between merit and connections, while others find complementarity. Together, this evidence suggests that the balance between merit and networks is not fixed but shaped by the broader institutional environment.

Since its inception in the early seventh century, and especially after its expansion in the eleventh, the imperial examination has been both recognized and debated for its meritocratic character. An extensive interdisciplinary literature argues that the system—through its formal openness, institutional design to recruit talent nationwide in place of aristocrats, and promotion of Confucian ideology that reinforced respect for state authority—shaped political power structures (e.g., Lv, 1944; Huang, 2023; Jia, Roland and Xie, 2024), enabled social mobility (e.g., Kracke, 1947; Ho, 1962; Jiang and Kung, 2021; Wen, Wang and Hout, 2024), forged cultural identity (e.g., Bol, 1994; Elman, 2000), and contributed to political stability (e.g., Bai and Jia, 2016; Huang and Yang, 2022).[1] Our results highlight two intertwined features of the system: its institutional design for meritocratic selection and its function as a crucible for network formation. As noted earlier, this

[1] Kung (2022) and Jia and Kung (2025) provide reviews of part of this literature, indicating that the literature is vast, but even these reviews cannot cover everything.

dual character appears to be a general feature of many meritocratic institutions across societies.

2 Background and Data

2.1 The Examination

The imperial examination system was established around 605 CE and expanded significantly during the Song dynasty around 1000 CE. It reached maturity and stability under the Ming and Qing dynasties, before being abolished in 1905 as China turned toward modern educational institutions.

In the Qing dynasty, the examination system operated at three levels: the entry (prefectural) level, held twice every three years; the provincial level, held once every three years; and the national (metropolitan) level, also held once every three years. In addition to these regular examinations, special sessions were occasionally held to commemorate significant imperial events or respond to exceptional circumstances.

Success at the national level qualified candidates for important offices. On average, about 240 candidates passed the national exam in each cohort. During our study period, there were 112 such exams, producing a total of 24,269 successful candidates from across the empire.

Candidates were ranked by their examination performance and further divided into three categories. The first consisted of the top three men, whose names were publicized nationwide. The second comprised a larger group of successful candidates who also enjoyed high prestige and strong career prospects. The third, the largest group, were more likely to secure positions in provincial rather than national offices. In Chinese, the first two categories were awarded the *Jinshi* degree, while the third received the *Tong Jinshi* degree (literally "similar to Jinshi"). For simplicity, we refer to the first two groups as the high (or better-performing) category and the third as the low (or worse-performing) category. Candidates in the high category not only held greater social standing but also typically began their official careers from more favorable positions.

Data and measurement We obtain examination performance data from the directory of the exam graduates (Zhu and Xie, 1980). To make results comparable across the 112 exam cohorts, we normalize performance using percentile ranks. Given the institutional division into high and low categories, we define percentiles relative to the cutoff that separates the two groups. In our analysis, we allow the possibility that performance mattered differently for these two categories.

2.2 Political Offices

Our focus is on understanding which candidates advanced to top office, defined as national-level offices, such as ministers, governors general, and provincial governors. In the long-standing nine-rank official system, these positions were ranked three or above. Our data on political offices and their ranks come from the *Chronicle of Officials in the Qing Dynasty* (Qian, 2005).

Although the examination system created a pool of talent for these positions, the probability of advancement remained low: Only about 12.6 percent of successful candidates eventually attained top office. On average, it took 16 years from passing the exam to reaching such positions. The historical data do not provide detailed career histories for each individual. Thus, we focus on whether a candidate could eventually make it to the top positions. Our data also allow us to take a closer look at the timing of achieving such positions.

Political appointments were formally made by the emperor, but in practice the process was shaped by key central figures, notably the Grand Secretaries of the Grand Secretariat (*Da Xueshi*) and the Grand Ministers of State (*Junji Dachen*). On average, seven individuals occupied these two offices in any given year. We focus on the ties to examiners who achieved such key central positions.

2.3 Nature and Exogeneity of the Examiner–Examinee Network

As the sole channel of elite recruitment, the examination system also became a locus of network formation among political elites. Ties forged through the examinations endured well beyond the testing halls, shaping the subsequent careers of aspiring officials. Historians have described the system as "a veritable faction factory" (Kuhn, 2003). Miyazaki (1981) vividly depicts these master–disciple relationships:

> The new graduates paid their respects to the examiners, whom they considered their lifelong teachers, and entered a firm master–disciple pledge. The chief and associate examiners were called "master teacher," and the assistant examiners "teacher," while the graduates called themselves "disciples" and referred to each other as "classmates." . . . The result was a pledge between them to assist each other to weather the storms of political life. (Miyazaki, 1981, p. 56)

Each national-level examination was overseen by one or two chief examiners and two or three vice examiners, all of whom were serving central government officials when appointed. It is

especially telling that newly selected candidates paid formal visits to the examiners only after the examination, since candidates were not permitted to know the examiners' identities in advance. At the same time, because examination papers were anonymized during the grading process, examiners did not know the identities of the selected examinees beforehand. These institutional features make examiner–examinee ties plausibly exogenous to individuals at the time they were formed.

Once the examination concluded, however, social norms encouraged the formation of enduring examiner–examinee factions. Because some examiners later rose to key positions in the central government, this setting provides an opportunity to study how the career trajectories of examiners shaped the prospects of their disciples.

By helping disciples "weather the storms of political life," these networks influenced policymaking (Kuhn, 2003) and at times played decisive roles in critical historical moments. For example, examination networks were mobilized during the suppression of the Taiping Rebellion, both to recruit military participation and to determine the postwar distribution of power (e.g., Platt, 2007; Bai, Jia and Yang, 2023). In this article, we focus on the master–disciple links and their effects on individual career outcomes.

Figure 1 depicts the links between examiners and examinees in the Qing dynasty, based on information from the records of the imperial examination (Jiang, Li and Chen, 2010) and the directory of the exam graduates (Zhu and Xie, 1980). Since the Qing dynasty was ruled by a small Manchu elite governing a large Han population and the examination system primarily selected Han elites, we also incorporate the ethnicity of examiners into our analysis. As the figure shows, although Manchus accounted for less than 1 percent of the population, they were disproportionately represented among examiners, serving in 15 of the 112 national exams. This overrepresentation reflects the political importance of examiner positions, which involved selecting the empire's future officials. In contrast, Han examiners supervised 80 exams, while 17 exams were jointly overseen by both Manchu and Han examiners.

To measure ties to powerful examiners, we proceed in three steps. First, we code whether examiner j in exam year t held a key central position in year $t+n$, denoted by $I_{j,t+n}$. Second, we sum across all J examiners in that cohort. Third, we aggregate over N years following exam year t to capture the duration of examiner power:

$$\text{Power}_c = \sum_{n=1}^{N} \sum_{j=1}^{J} I_{j,t+n}. \tag{1}$$

In the baseline, we set $N = 10$ and use $N = 5$ and $N = 20$ as robustness checks. We also exploit examiners' pre-exam success in $t - 10$ (and $t - 5$, $t - 20$) to aid interpretation of the influence of power. Moreover, we leverage the power trajectories of examiners to address potential concerns of reverse causality.

3 Research Design

We begin with the following specification, which accounts for the institutional cutoff separating the high and low categories:

$$\begin{aligned} Y_i = \alpha + \beta_1 \text{Perf}_i + \beta_2 \text{High}_i + \beta_3 \text{High}_i \times \text{Perf}_i \\ + \beta_4 \text{Power}_c + X_i' \gamma + \varepsilon_i, \end{aligned} \tag{2}$$

where Y_i is a dummy variable indicating whether *Jinshi i* eventually became a top (i.e., national-level) official. Perf_i denotes exam performance, measured in every 10 percentile points from the cutoff that separates the low and high categories (denoted by High_i). The institutional design distinguishing higher from lower categories implies that those in the high category began with a more favorable position and were more likely to advance further. To account for this, we include the interaction term $\text{High}_i \times \text{Perf}_i$.

Power_c is measured by equation (1), where $N = 10$ is our baseline. $I_{j,t+n} = 1$ if examiner j held a Grand Secretary or Grand Minister within 10 years after the exam year and 0 otherwise. We use $N = 5$ and $N = 20$ as robustness checks and also include pre-exam power to distinguish a patronage interpretation from alternatives such as mentoring or human capital transmission.

X_i is a vector of controls that includes the ethnicity of examiners (measured by the share of Manchu examiners in a cohort), cohort size (number of *Jinshi* in a cohort), examinees' hometown province dummies, and prefecture-level characteristics including logged quotas at entry level, logged total population, and logged area. We also conduct specifications with hometown prefecture fixed effects, which compare examinees from the same prefecture.

Our interest lies in assessing how the estimates change with and without these regional background controls. If the coefficients were to change substantially, it would suggest that personal background serves as an important confounder. However, we find that the results remain stable, indicating that this is not the case. Although we lack direct measures of family background, the inclusion of hometown characteristics provides a useful robustness check.

Perf_i varies at the individual level and Power_c varies at the cohort level. Because variation across exam cohorts is central to our analysis and because individuals from the same cohort may share common examiners, institutional conditions, or shocks, we allow for potential correlations within cohorts by clustering standard errors at the cohort level.

Beyond this baseline, our second set of analyses examines how the effect of examiner power varies with exam performance, and our third analysis explores how the relative importance of performance and examiner power changes over time.

4 Results

4.1 Importance of Exam Performance

As motivating evidence, Figure 2 plots the correlation between exam performance and the probability of becoming a top official. The figure shows a clear positive relationship. In addition, the institutional cutoff separating the high and low categories also matters: The probability of success exhibits a noticeable jump at the cutoff, and the performance–success slope becomes higher for the high category.

To better understand the magnitude, we present the estimates in Table 1. Column (1) shows the average association between exam performance and success. On average, a 10 percentile increase in exam rank is associated with a 3 percentage point higher likelihood of becoming a top official, equivalent to 23.8 percent of the mean.

Institutional thresholds also mattered. Column (2) of Table 1 and Figure 2 show a discontinuous jump in the probability of becoming a top official at the cutoff, with a magnitude of 1.9 percentage points (15 percent of the mean). This pattern suggests that, beyond continuous performance differences, the formal categorization itself independently influenced political selection. Although candidates close to the cutoff on either side of the cutoff may have been equally capable, those above it were institutionally designated for superior career prospects.

The strength of the performance–career link further differed by category. For high-category candidates, a 10 percentile increase in rank is associated with a 6.6 percentage point higher likelihood of reaching top office, compared with only 0.7 percentage points among low-category candidates.

Taken together, these findings are consistent with the system's meritocratic features: Exam

performance strongly predicted subsequent career success, and the institutional thresholds that formally distinguished high from low categories further reinforced this link. The discontinuous jump at the cutoff highlights that institutional classifications were not merely symbolic: They are associated with materially different career trajectories. Candidates just above the threshold, though only marginally higher in performance, enjoyed a meaningful advantage in attaining top office.

The steeper performance–success gradient is also informative. Those officially recognized as belonging to the higher tier benefited not only from status but also from enhanced visibility and access to opportunities, allowing performance to translate more effectively into advancement.

Overall, these patterns reveal that while the imperial examination system did reward exam performance, it did so through a combination of measured merit (however imperfect) and categorical recognition. Meritocracy, in this sense, functioned both as an assessment of individual merit and as an institutional hierarchy that formally structured the pathways to elite status.

4.2 Influence of Examiner Power

Examiner power also mattered. We plot the raw correlation between examiner power and the likelihood of achieving top office in Figure 3, which is clearly positive.

In columns (3)–(4) of Table 1, we compare exam performance and examiner power. A one-standard-deviation increase in such ties is associated with a 2.1 percentage points (17 percent) higher probability of achieving top office. This relationship is comparable in magnitude to that of a 7 percentile increase in exam performance.

In columns (5)–(7), we further control for examiner ethnicity, prefecture characteristics, and even prefecture fixed effects. The coefficients are stable across specifications, indicating that our measures of exam performance and ties to powerful examiners are unlikely to reflect omitted personal background characteristics. The positive association between ties to powerful examiners and career success, however, does not by itself demonstrate the causal influence of networks. In what follows, we consider two alternative interpretations: first, whether the relationship reflects human capital transmission from examiners to examinees, and second, whether it arises from reverse causality, with examiners and their former examinees advancing together and helping each other.

Networks versus human capital A plausible alternative explanation is that more successful examiners were also better mentors who transmitted additional human capital to their students.

To distinguish between these interpretations, we exploit time-varying information on examiners' career trajectories. Specifically, we conduct a horse race test comparing examiners' achievements in the 5, 10, and 20 years before and after examiner–examinee ties were formed.

If our results primarily capture network influence and access to power, we expect examiners' post-exam power to matter more, since their pre-exam power would have gradually waned over time. Conversely, if the effect operates through mentoring or human-capital transmission, the timing of examiners' success should be less relevant, and pre-exam power should be equally important.

The results, shown in columns (1)–(2) of Table 2, indicate that post-exam power dominates, and including pre-exam power alters the estimates little. Columns (3)–(6), which extend the power window from 5 to 10 and 20 years, yield highly similar results. These patterns are more consistent with network access than with a pure mentoring channel, though the two channels are not fully separable. We do not rule out the possibility that powerful examiners also mentored their examinees in the political process. Still, the evidence suggests that examiner power itself was an important part of what made these ties consequential for the examinee's political success.

Examining reverse causality Another potential concern in interpreting the correlation between examiner power and the subsequent career success of examinees is reverse causality. It is possible that examiners' own success was influenced by their examinees or that both groups advanced simultaneously along similar career trajectories.

Ideally, we would address this by conducting an event study that examines examinees' outcomes following a loss of examiner power. Unfortunately, our career data are not sufficiently detailed to allow for such an analysis.

To probe this possibility, we exploit variation in examiners' career trajectories around the time of the exam. For each examiner, we measure changes in power in the years before and after the exam and classify them into two groups: (i) those on a rising trajectory, whose power increased after the exam, and (ii) those on a nonrising trajectory, whose power remained stable or declined. We then reestimate our baseline specification separately for these two groups.

If reverse causality were driving the results, i.e., examiners' advancement was partly fueled by the subsequent success of their examinees, we would expect stronger effects among the rising-trajectory examiners, as both examiners and examinees would have advanced together. In contrast, if the results primarily reflect network influence where examiners' later power benefited examinees through power in position, we should observe similar effects across the two groups.

As shown in Table 3, the estimated coefficients are remarkably similar for both rising- and nonrising-trajectory examiners. This symmetry indicates that simultaneous career advancement is unlikely to explain our findings and supports the interpretation that examiner power affected examinee outcomes.

Merit versus networks over the career cycle We further assess the roles of merit and examiner power by examining the timing of appointments to top office. Specifically, we consider whether an examinee entered top office within 0–9, 10–14, 15–19, 20–24, 25–29, or more than 30 years after the exam.

Figure 4 plots the estimated effects of exam performance and ties to powerful examiners across these career stages, alongside the mean probability of attaining top office. The probability follows a hump-shaped pattern, peaking in mid-career. Notably, the impact of examiner power emerges already in the first 0–9 years after the exam, revealing their importance in shaping the early stages of bureaucratic advancement. This effect is particularly remarkable given that, during this period, most candidates were only beginning their official career and few had yet reached top office.

Comparison with other networks We focus on examiner–examinee ties in this study, which were formally established through the examination system. Yet these were by no means the only networks that individuals sought to build and sustain. To provide a comparison, we also examine hometown ties, measured by the number of key officials (Grand Secretariat and the Grand Ministers of State) from the same province as an examinee within five years after the exam.

As shown in column (1) of Table 4, both types of connections were associated with improved career outcomes, but the effects of examiner power are substantially stronger. When standardized by a one-standard-deviation change in each measure, the influence of examiner power is 4.25 times that of hometown ties. This pattern suggests that networks forged through the examination system were more cohesive and functioned as well-defined factions within imperial politics.

Historical research has long noted that emperors were acutely aware of these alliances and repeatedly attempted to dilute their influence. One reason the emperor personally ranked the finalists in the palace examination was precisely to assert imperial authority and legitimacy in the process of conferring office upon new elites. Nevertheless, examiners occupied a special position within this hierarchy, as they directly determined who could be selected through the examination.

In column (2), we examine the interaction between examiner power and hometown ties and find no evidence of a significant interaction between them. This result suggests that these two forms of

connections contribute jointly to officials' career advancement, yet neither crowd in nor crowd out the other.

4.3 The Interplay between Merit and Networks

Strong complementarity We study the interplay between merit and networks by estimating the effects of exam performance and ties to powerful examiners separately for the high- and low-performing categories. Figure 5 plots the corresponding coefficients: The left panel shows the effects of exam performance for the two categories, while the right panel presents those of examiner power.

Two patterns stand out. First, exam performance matters more for the high category, consistent with the institutional design that steered this elite group toward more prestigious careers. Second, ties to powerful examiners are also more valuable for the high category, pointing to a complementarity between merit and networks. The same degree of examiner power is nearly twice as beneficial for candidates in the high category as for those in the low category.

To study the complementarity more systematically, we extend our baseline analysis by adding an interaction term between exam performance and examiner power. We measure exam performance in two ways: (1) a dummy variable distinguishing high and low categories, and (2) a continuous measure based on exam percentiles. As reported in Table 5, the interaction between exam performance and examiner power is strongly positive under both specifications, providing robust evidence of complementarity. Quantitatively, given the same level of examiner power, the additional advantage associated with being in the high-performing category is comparable to that of a one-standard-deviation increase in examiner power.

A counterfactual of substitution To understand the implications of our finding on complementarity, it is useful to contrast it with the counterfactual of substitution. When merit and networks are complements (rather than substitutes), those who attained top office tend to exhibit both stronger exam performance and stronger connections. By contrast, under substitution, strong networks would compensate for weaker exam performance, and vice versa.

This distinction has two key implications. First, complementarity implies that the degree of meritocracy is arguably higher than it would be under substitution, since successful candidates appear to have had better performance. Second, complementarity makes it harder to identify the influence of networks, because their effects can operate under the guise of merit rather than as a

visible substitute for it.

4.4 Persistence and Change Amid Dramatic Political Shifts

The long-term horizon of our data allows us to examine how the relative importance of merit and networks evolved alongside broader institutional changes. Because our main variation in networks arises across exam cohorts, we divide the Qing dynasty into two periods. A natural historical breakpoint is 1839–1840, marking the onset of the Opium War and the ensuing era of internal unrest and external shocks. This division allows us to ask whether the influence of merit and networks shifted in response to these broader political transformations.

Enduring influence of exam performance and examiner power Figure 6(a) plots four sets of coefficients, distinguishing high versus low categories, for cohorts before and after 1840. The results highlight the persistence of exam performance as a determinant of political advancement. Across both periods, the estimated coefficients remain stable and significant, suggesting that, despite the political disruptions and shifting institutional environments, candidates' exam achievements continued to serve as a signal of merit in official selection.

Figure 6(b) presents analogous estimates for examiner power. A similar pattern emerges: Ties to powerful examiners consistently enhanced examinees' prospects throughout the Qing dynasty. Even after 1840, when foreign incursions and internal crises began reshaping the country, connections to influential examiners retained their value. Together, these findings reveal the enduring roles of both merit and networks in shaping elite formation and suggest that the dual pillars of exam performance and examiner influence remained resilient even amid dramatic political and institutional transformations.

Change in the weight of examiners Beyond the persistence of examiner influence, we also detect shifts in which type of examiner power carried more weight. A central dimension concerns the ethnicity of examiners. Existing scholarship notes that the Qing rulers increasingly relied on Han elites in the later periods (e.g., Twitchett and Fairbank, 1978; Bai, Jia and Yang, 2023). This provides an opportunity to assess whether such shifts in the broader political balance are reflected in the selection of officials through the examination system.

To investigate this, we decompose an examinee's ties to powerful examiners into two categories: ties to powerful Manchu examiners and ties to powerful Han examiners. We then estimate their influence separately. As reported in Table 6, both types of power mattered, but their relative

importance changed over time. In particular, the estimated influence of Manchu examiner power fell sharply after the 1840s, declining from 0.016 and statistically significant to 0.008 and no longer statistically distinguishable from zero. By contrast, the estimated influence of Han examiner power rose by more than 50 percent, from 0.015 to 0.024. These patterns mirror the gradual rebalancing of authority within the imperial government.

These results point to both persistence and change. What persisted was the overarching importance of examiner power in shaping careers. What changed was the identity of the examiners whose power mattered most. This pattern highlights the resilience of network influence, even as the underlying political order underwent dramatic transformations.

5 Discussion and Conclusion

Across societies, institutions that claim to allocate opportunity on the basis of merit—whether through examinations, elite schools, or bureaucratic selection systems—are rarely insulated from the formation and influence of social networks. Relationships between mentors and protégés, or among peers, consistently shape elite careers. The persistence of these informal forces has led scholars across disciplines to revisit Young's (Young, 1962) satirical depiction of meritocracy and to question its very existence (e.g., Rivera, 2016; Markovits, 2019; Sandel, 2021).

This skepticism is also reflected in how survey instruments are designed. Influential surveys that measure beliefs in meritocracy often build on the assumption that merit and patronage are substitutes. Consider, for example, the World Values Survey item that measures beliefs about meritocracy. Asking respondents to place themselves on a 0–10 scale, the item ranges from "In the long run, hard work usually brings a better life" (0) to "Hard work doesn't generally bring success—it's more a matter of luck and connections" (10).

Our results suggest that meritocracy does not operate in a way that is dichotomous, but that the influence of individual accomplishment and social connection vary by degree in different settings. Success depends not only on measurable achievement but also on hidden forces that can appear to operate through merit. We show that merit and networks in the Qing examination system were not substitutes but complements. This distinction matters. Complementarity means that networks did not merely override performance but instead enhanced the visibility and rewards of high performers. In this sense, the system was arguably more meritocratic than one in which networks crowd out merit altogether. Yet the complementarity also has a darker side: It can legitimize patronage by

cloaking it in the language of merit. Candidates connected to powerful examiners could plausibly present their advancement as deserved, even when networks were pivotal.

Specific to China, the imperial examination system has long been celebrated as a quintessentially meritocratic institution, with profound implications for the Chinese state and society. Our findings complicate this view. They show that the system functioned not only as a mechanism to identify talent but also as a crucible for forging networks that shaped political careers. These dynamics reveal the resilience of informal forces even within one of history's most enduring meritocratic institutions.

More broadly, the lessons extend beyond China. Modern education and selection systems continue to face similar tensions between formal achievement and informal connections. Elite universities, professional gatekeepers, and bureaucratic hierarchies all blend merit with networks in ways that affect perceptions of fairness and legitimacy. The Chinese case reminds us that these tensions are not new but deeply rooted in the way institutions structure opportunity. Understanding how merit and networks interact, whether they substitute or complement, remains central for assessing the promise and limits of meritocracy today.

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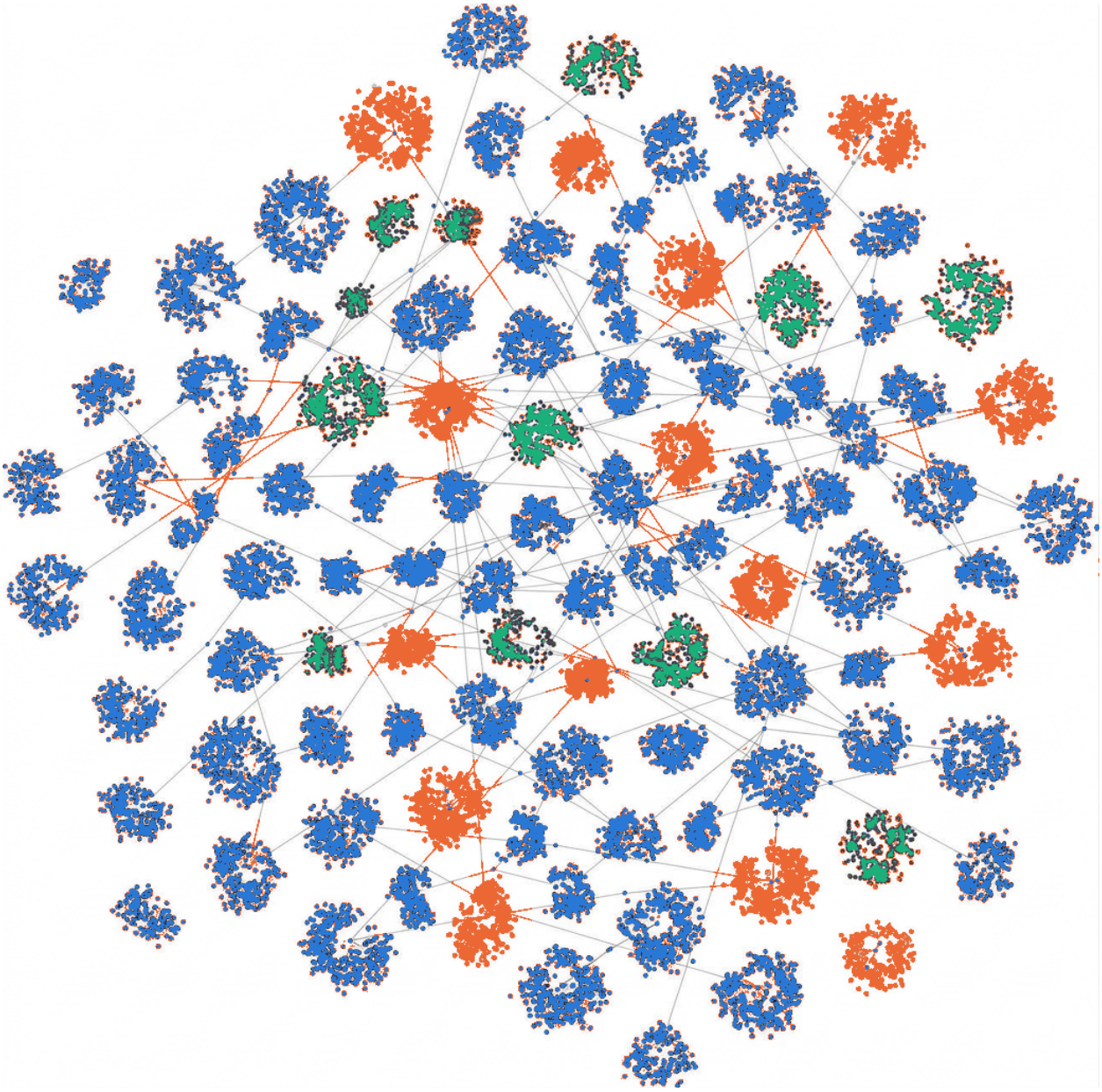
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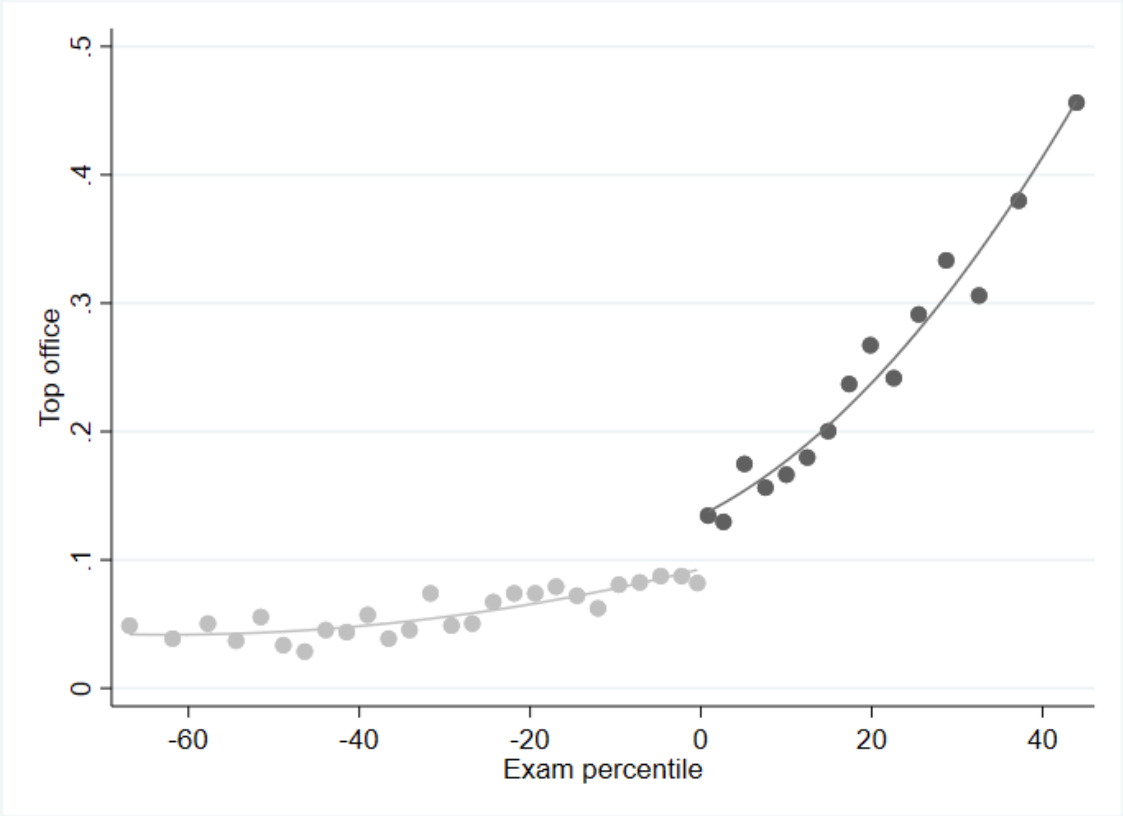
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Figure 1: The Examiner–Examinee Network



NOTE: The figure is made by the authors based on data from Zhu and Xie (34). Between 1644 and 1905, Qing China held 112 exams, which recruited a total of 24,269 *Jinshi*. Blue dots denote examinations with Han examiners, orange dots denote those with Manchu examiners, and green dots denote those jointly overseen by Han and Manchu examiners.

Figure 2: Exam Performance and the Probability of Attaining Top Office



NOTE: The x-axis shows candidates' exam performance, measured in percentile points relative to the cutoff that divides the high category (*Jinshi*) from the low category (*Tong Jinshi*); positive values indicate better performance within the high category and negative values indicate lower performance within the low category. The y-axis shows the probability of eventually attaining a top office (a national-level position ranked 3 or above). Each point represents the mean outcome within a percentile-point bin.

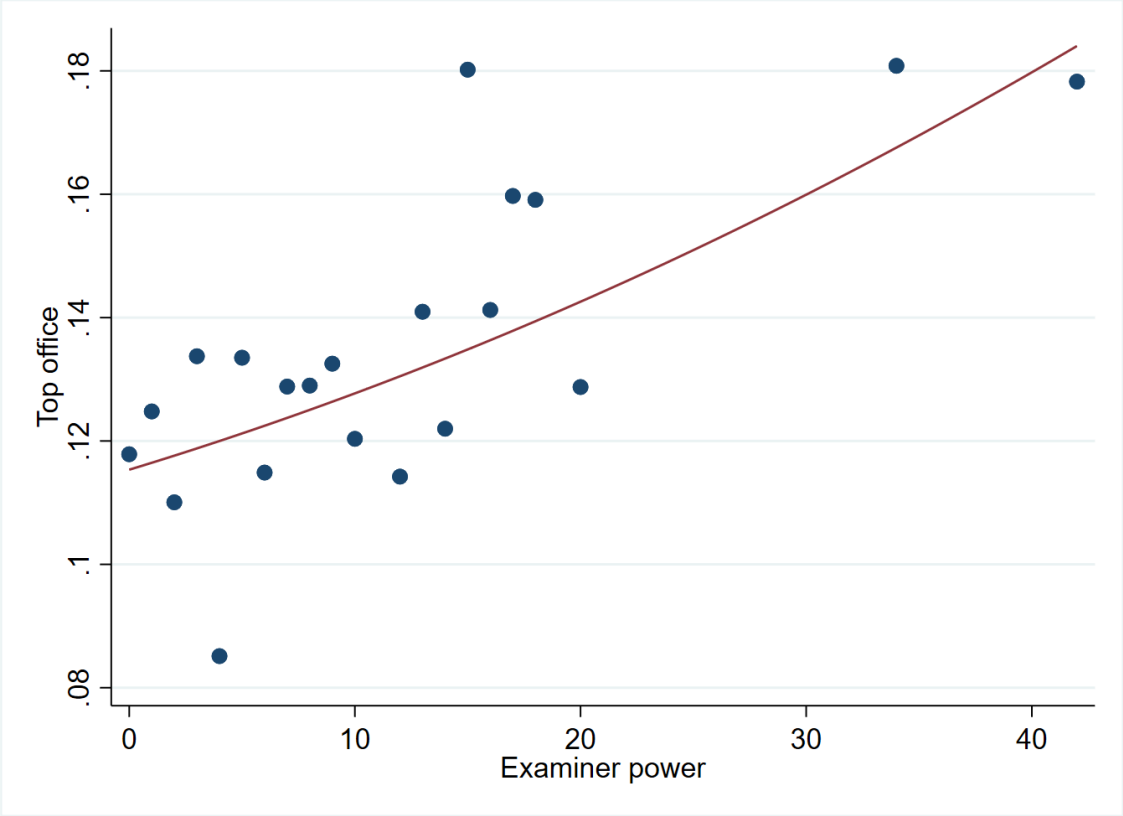
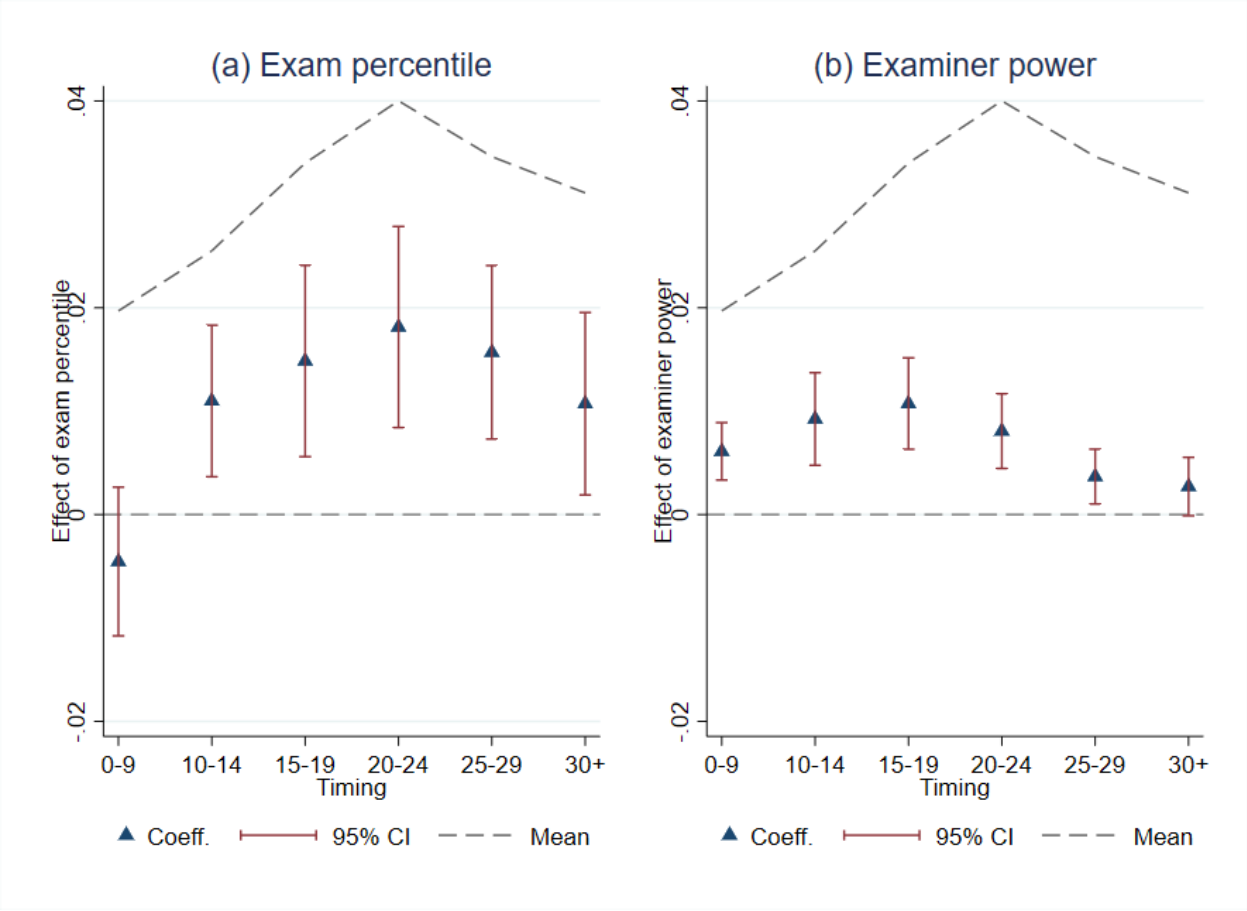


Figure 3: Examiner Power and Attaining Top Office

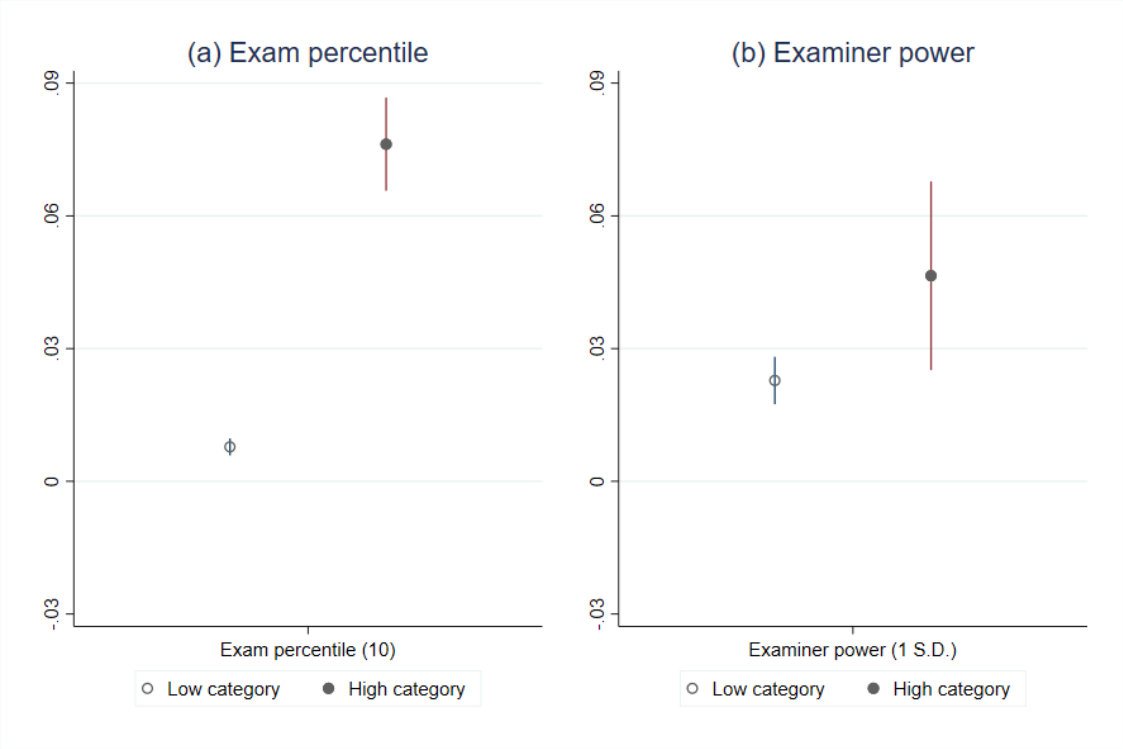
NOTE: The x-axis shows candidates' ties to powerful examiners, measured by the total number of years that the cohort's examiners held key central posts (Grand Secretariat or Grand Ministers of State) within 10 years after the exam. The y-axis shows the probability of eventually attaining a top office (a national-level position ranked 3 or above). Each point represents the mean outcome within a bin of the examiner power measure.

Figure 4: Merit Versus Ties to Powerful Examiners in One's Career Cycle



NOTE: We generate a set of binary indicators for whether a *Jinshi* first attained a top office within 0–9, 10–14, 15–19, 20–24, 25–29, or 30 or more years after passing the exam. Each indicator is regressed separately on exam percentile, the high-category dummy, their interaction, and examiner power, controlling for Manchu examiner share, cohort size, and hometown prefecture fixed effects. Panel (a) plots the estimated effect of a 10 percentile point increase in exam rank. Panel (b) plots the estimated effect of a one-standard-deviation increase in examiner power. Triangles show point estimates; bars show 95 percent confidence intervals. The dashed line shows the unconditional mean probability of attaining top office during the corresponding career window (i.e., first appointment to top office occurring within that period). Examiner ties emerge as particularly influential early in the career cycle, when overall advancement rates are still low and few candidates have yet reached top office.

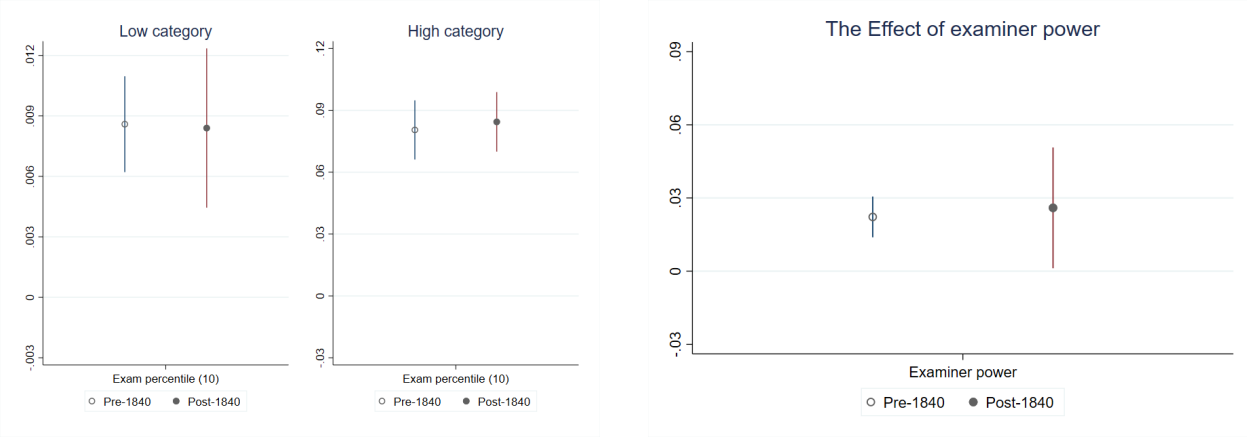
Figure 5: Exam Performance Versus Examiner Power by Category



NOTE: We divide the sample into the low category (*Tong Jinshi*) and the high category (*Jinshi*) and estimate separate regressions for each subsample. Each regression includes exam percentile, examiner power, Manchu examiner share, cohort size, and hometown prefecture fixed effects. The high-category dummy and its interaction with exam percentile are not included within each subsample regression, as the sample is already restricted by category. Panel (a) plots the coefficients on exam percentile (10 percentile point increase) for the two subsamples. Panel (b) plots the coefficients on examiner power (one-standard-deviation increase). Empty dots represent the low category; solid dots represent the high category. Bars represent 95 percent confidence intervals.

Figure 6: Roles of Exam Performance and Examiner Power, Pre- and Post-1840

(a) Merit before and after 1840 **(b)** Examiner power before and after 1840



NOTE: We divide the sample into pre-1840 cohorts and post-1840 cohorts, corresponding to the onset of the Opium War and the ensuing period of political crisis. We estimate regressions for each subsample including exam percentile, the high-category dummy, their interaction, and examiner power, with Manchu examiner share, cohort size, and hometown prefecture fixed effects. Panel (a) plots the estimated effects of exam performance: the coefficient on exam percentile (10 percentile point increase) for the low category, and the total marginal effect for the high category. Panel (b) plots the estimated effects of examiner power (one-standard-deviation increase) for each subsample. Empty dots represent pre-1840 cohorts; solid dots represent post-1840 cohorts. Bars represent 95 percent confidence intervals.

Table 1: Exam Performance Versus Ties to Powerful Examiners in Attaining Top Office

	Top Office (Mean: 0.126)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Exam percentile (10)	0.0296***	0.0069***	0.0304***	0.0080***	0.0080***	0.0080***	0.0077***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
High category		0.0187*		0.0156	0.0162	0.0160	0.0155
		(0.011)		(0.010)	(0.010)	(0.010)	(0.010)
High category × Exam pctile (10)		0.0664***		0.0671***	0.0670***	0.0670***	0.0675***
		(0.006)		(0.006)	(0.006)	(0.006)	(0.005)
Examiner power (1σ)			0.0213***	0.0215***	0.0298***	0.0297***	0.0295***
			(0.003)	(0.003)	(0.004)	(0.004)	(0.004)
Observations	24,269	24,269	24,269	24,269	24,269	24,269	24,259
R-squared	0.0852	0.1083	0.0892	0.1123	0.1170	0.1172	0.1257
Province FE	Y	Y	Y	Y	Y	Y	N
Manchu examiner share	N	N	N	N	Y	Y	Y
Cohort size	N	N	N	N	Y	Y	Y
Prefecture-level controls	N	N	N	N	N	Y	N
Prefecture FE	N	N	N	N	N	N	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above (minister, governor general, or provincial governor); 0 otherwise. Exam percentile (10) denotes a 10 percentile point increase in exam rank, measured relative to the cutoff that divides the high and low category. High category = 1 for candidates who placed in the first or second group (awarded the *Jinshi* degree); the low category (*Tong Jinshi*) is the omitted group. Examiner power (1σ) is the total number of years that the cohort's examiners held key central posts (Grand Secretariat or Grand Ministers of State) within 10 years after the exam, standardized by one standard deviation. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi* in the cohort. Prefecture-level controls include logged entry-level quotas, logged total population, and logged area. Columns (1)–(6) include hometown province fixed effects; column (7) replaces these with hometown prefecture fixed effects, comparing candidates from the same prefecture, where 10 observations are absorbed due to the inclusion of prefecture fixed effects. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 2: Examiner Power: Networks Versus Human Capital

	Top Office (Mean: 0.126)					
	(1)	(2)	(3)	(4)	(5)	(6)
	5 Years		10 Years		20 Years	
Examiner power, post-exam (1σ)	0.0275***	0.0269***	0.0295***	0.0322***	0.0253***	0.0256***
	(0.003)	(0.007)	(0.004)	(0.006)	(0.005)	(0.006)
Examiner power, pre-exam (1σ)		0.0008		−0.0039		−0.0006
		(0.007)		(0.005)		(0.005)
Exam percentile (10)	0.0075***	0.0075***	0.0077***	0.0077***	0.0074***	0.0074***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
High category	0.0161	0.0161	0.0155	0.0153	0.0161	0.0160
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
High cat. × Exam pctile (10)	0.0673***	0.0673***	0.0675***	0.0676***	0.0675***	0.0675***
	(0.006)	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)
Observations	24,259	24,259	24,259	24,259	24,259	24,259
R-squared	0.1249	0.1249	0.1257	0.1258	0.1244	0.1244
Manchu examiner share	Y	Y	Y	Y	Y	Y
Cohort size	Y	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above; 0 otherwise. Exam percentile (10) denotes a 10 percentile point increase in exam rank relative to the category cutoff. Examiner power, post-exam (1σ) is the total years examiners held key central posts (Grand Secretariat or Grand Ministers of State) in a window after the exam, standardized by one standard deviation. Examiner power, pre-exam (1σ) is the analogous measure for the symmetric window before the exam. Columns (1)–(2) use a 5-year window (post-exam: years t to $t+4$; pre-exam: years $t-5$ to $t-1$). Columns (3)–(4) use a 10-year window. Columns (5)–(6) use a 20-year window. All columns include prefecture fixed effects, and 10 observations are absorbed due to the inclusion of prefecture fixed effects. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi*. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 3: Testing Possible Reverse Causality

	Top Office (Mean: 0.126)	
	(1)	(2)
Examiner types	Power increased	Power decreased or unchanged
Examiner power, post-exam (1σ)	0.0300***	0.0319***
	(0.006)	(0.006)
Exam percentile (10)	0.0089***	0.0071***
	(0.002)	(0.001)
High category	0.0301**	0.0015
	(0.014)	(0.014)
High category × Exam percentile (10)	0.0590***	0.0756***
	(0.009)	(0.006)
Observations	11,341	12,897
R-squared	0.1258	0.1436
Manchu examiner	Y	Y
Cohort size	Y	Y
Prefecture FE	Y	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above; 0 otherwise. Exam percentile (10) denotes a 10-percentile-point increase in exam rank relative to the category cutoff. Examiner power, post-exam (1σ) is the total years examiners held key central posts within 10 years after the exam, standardized by one standard deviation. Column (1) restricts the sample to cohorts whose examiners were on a rising trajectory. This cohort includes 11,357 *Jinshi*, 16 of whom are absorbed due to the inclusion of prefecture fixed effects. Column (2) restricts the sample to cohorts whose examiners were on a non-rising trajectory (stable or declining). This cohort includes 12,912 *Jinshi*, 15 of whom are absorbed due to the inclusion of prefecture fixed effects. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi*. Both columns include prefecture fixed effects. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 4: Hometown Ties as a Comparison

	(1)	(2)
	Top Office (mean: 0.126)	
Exam percentile (10)	0.0077***	0.0077***
	(0.001)	(0.001)
High category	0.0151	0.0151
	(0.010)	(0.010)
High category × Exam percentile (10)	0.0676***	0.0676***
	(0.005)	(0.005)
Examiner power (1σ)	0.0285***	0.0263***
	(0.004)	(0.006)
Hometown key officials (1σ)	0.0067**	0.0062*
	(0.003)	(0.003)
Examiner power × Hometown key officials (1σ)		0.0020
		(0.003)
Observations	24,259	24,259
R-squared	0.1260	0.1261
Province FE	Y	Y
Manchu examiner	Y	Y
Cohort size	Y	Y
Prefecture FE	Y	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above; 0 otherwise. Exam percentile (10) denotes a 10 percentile point increase in exam rank relative to the category cutoff. Examiner power (1σ) is the total years the cohort's examiners held key central posts within 10 years after the exam, standardized by one standard deviation. Hometown key official power (1σ) is the number of Grand Secretariat members and Grand Ministers of State from the candidate's home province who held office within five years after the exam, standardized by one standard deviation. Column (1) includes both examiner power and hometown ties. Column (2) adds their interaction. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi*. Both columns include prefecture fixed effects, and 10 observations are absorbed due to the inclusion of prefecture fixed effects. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Complementarity Between Exam Performance and Examiner Power

	Top Office (Mean: 0.126)	
	(1)	(2)
High cat. × Examiner power (1σ)	0.0220**	
	(0.010)	
Exam pctile × Examiner power (1σ)		0.0028**
		(0.001)
Examiner power (1σ)	0.0236***	0.0318***
	(0.003)	(0.005)
Exam percentile (10)	0.0074***	0.0045***
	(0.001)	(0.002)
High category	−0.0048	0.0158
	(0.012)	(0.010)
High cat. × Exam pctile (10)	0.0692***	0.0695***
	(0.005)	(0.005)
Observations	24,259	24,259
R-squared	0.1266	0.1263
Manchu examiner share	Y	Y
Cohort size	Y	Y
Prefecture FE	Y	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above; 0 otherwise. Examiner power (1σ) is the total years examiners held key central posts within 10 years after the exam, standardized by one standard deviation. Column (1) measures exam performance using the High category dummy (= 1 for *Jinshi*, 0 for *Tong Jinshi*) and includes its interaction with examiner power. Column (2) measures exam performance using the continuous Exam percentile (10), denoting a 10 percentile point increase in exam rank relative to the category cutoff, and includes its interaction with examiner power. All columns include hometown prefecture fixed effects, and 10 observations are absorbed due to the inclusion of prefecture fixed effects. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi*. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Change in the Weight of Examiners

	Top Office (mean: 0.126)				
	(1) Pre 1840	(2) Post 1840	(3) All	(4) Pre 1840	(5) Post 1840
Examiner power (1σ)	0.0222***	0.0260**			
	(0.004)	(0.012)			
Manchu examiner power (1σ)			0.0159***	0.0162**	0.0080
			(0.005)	(0.006)	(0.008)
Han examiner power (1σ)			0.0229***	0.0153***	0.0235**
			(0.004)	(0.004)	(0.010)
Exam percentile (10)	0.0086***	0.0084***	0.0077***	0.0086***	0.0084***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
High category	0.0294**	−0.0434***	0.0157	0.0299**	−0.0434***
	(0.013)	(0.013)	(0.010)	(0.013)	(0.013)
High cat. × Exam pctile (10)	0.0718***	0.0761***	0.0674***	0.0715***	0.0760***
	(0.007)	(0.007)	(0.006)	(0.007)	(0.007)
Observations	17,148	7,100	24,259	17,148	7,100
R-squared	0.1215	0.1863	0.1259	0.1220	0.1863
Manchu examiner share	Y	Y	Y	Y	Y
Cohort size	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y

NOTE: Top Office = 1 if the candidate ever attained a national-level office ranked 3 or above; 0 otherwise. Exam percentile (10) denotes a 10 percentile point increase in exam rank relative to the category cutoff. Examiner power (1σ) in columns (1)–(2) is the total years all examiners (regardless of ethnicity) held key central posts within 10 years after the exam, standardized by one standard deviation. Manchu examiner power (1σ) and Han examiner power (1σ) decompose the overall measure by the ethnicity of the examiners, each standardized separately. All columns include prefecture fixed effects. Columns (1)–(2) report overall examiner power for pre- and post-1840 cohorts. Column (3) decomposes examiner power by ethnicity for the full sample. Columns (4)–(5) repeat the decomposition for pre-1840 and post-1840 cohorts. Pre-1840 cohorts include 17,159 *Jinshi*, 11 of whom are absorbed due to prefecture fixed effects; post-1840 cohorts include 7,110 *Jinshi*, 10 of whom are absorbed. Manchu examiner share is the proportion of Manchu examiners in the cohort. Cohort size is the logged number of *Jinshi*. Standard errors clustered at the cohort level are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.